Kaiyue Chen

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EDUCATION

BEng in Mechatronics and Robotic Systems Sep 2024 – Jul 2028 (expected)
Xi'an Jiaotong-Liverpool University, Suzhou, China

- GPA (Years 1-2): 3.83/4.0

PUBLICATIONS

*Equal contribution.

- **TacFRS: Plug-in Flow Reversal for High-Frequency Tactile Revision of VLA Models**.
  Under review at the International Conference on Learning Representations (ICLR 2027).
  Chaoyi Liu*, **Kaiyue Chen***, Junlai Li*, Rongxuan Zhang, Tianjian Xu, Qin Huang, Hao Su, Rui Chen, and Jing Xu.
- **3WD-VLA: Hierarchical Three-Way Decision for Collaborative Dual-View Robotic Grasping**.
  Accepted at EAI CollaborateCom 2026, 22nd EAI International Conference on Collaborative Computing: Networking, Applications and Worksharing.
  Yan Zhu*, **Kaiyue Chen***, Lingyu Li, Xiaohui Zhu, and Xinheng Wang.

RESEARCH EXPERIENCE

Embodied Foundation Models and World Action Models, Research Intern Sep 2026 – Present
Peking Unversity, advised by Prof. Shanghang Zhang

- Contribute to a multimodal robot data processing platform, building a unified data adapter and trajectory indexing layer for DROID, AgiBot World Alpha, and RoboMIND, with on-demand access to TFRecord/RLDS, HDF5, Parquet, and compressed archives.
- Integrate and adapt 24 action quality checks and 7 video quality checks covering trajectory structure, kinematic consistency, abrupt action changes, inactivity, exposure, blur, and frozen frames; implement threshold calibration, anomaly localization, and manual review workflows.
- Implement rule-based text cleaning, automatic annotation with video models, episode-level deduplication, and reviewed data export, supporting concurrent inference, resumable processing, progress monitoring, and provenance tracking.
- Integrate and deploy NVIDIA FLORA's TempSeg, Captioner, and Rewarder inference code, including version pinning, SHA-256 integrity verification, bundled distribution, and isolated inference environments; develop a video temporal segmentation CLI with input validation, batch processing, and result auditing, alongside command-line interfaces for Captioner and Rewarder.
- Set up training and evaluation pipelines for $\pi_{0.5}$ and FastWAM, performed full-parameter fine-tuning, and evaluated both models across all tasks in the DexJoCo benchmark

TacFRS: Tactile Flow Reversal Steering, Co-first Author Jun 2026 – Sep 2026
Sudo Robotics, advised by Prof. Rui Chen

- Developed TacFRS, a plug-in framework for high-frequency tactile action revision of frozen flow-based VLAs, using finite-step flow reversal to encode action chunks into observation-conditioned structured noise.
- Built a tactile-conditioned flow decoder with temporal tactile encoding and proprioception, enabling 30 Hz action correction between low-frequency VLA updates.
- Designed tactile-gated supervision with flow-matching and endpoint reconstruction losses to preserve base-policy behavior while learning contact-dependent corrections.
- Evaluated SmolVLA and $\pi_{0.5}$ on four real-world single-arm and bimanual tasks; improved average success from 28.75% to 41.25% and from 53.75% to 67.50%, respectively.

ManiSkill-ViTac Challenge 2026, Student Organizer Mar 2026 – Jun 2026
CVPR 2026 Embodied AI Workshop: real-world language-guided bimanual vision-tactile manipulation

- Built an end-to-end VLA pipeline around UMI-based data collection, covering hardware maintenance, data processing and filtering, model training, and real-robot deployment.
- Integrated AnyTouch tactile representations into a $\pi_{0.5}$-based VLA using modality-weighted training, establishing the competition baseline.
- Designed two long-horizon, contact-rich tasks for model training and online deployment.
- Co-organized the challenge and contributed to the accompanying technical report.

Three-Way Decision VLA, Co-first Author Jan 2026 – May 2026
Xi'an Jiaotong-Liverpool University, advised by Prof. Xinheng Wang

- Developed a collaborative robotic grasping framework for cases where direct execution is unsafe or unreliable, including ambiguous language commands, target absence, human-hand intrusion, occlusion, and stacked objects.
- Designed a three-way decision framework for VLA policies to defer actions in ambiguous scenarios.
- Built a teleoperation data-collection pipeline for the SO-100 robot arm and trained ACT, $\pi_{0.5}$, and SmolVLA models.
- Improved performance by 27.5 percentage points over a matched SmolVLA baseline without 3WD and demonstrated clearer behavior under ambiguity, occlusion, and object stacking.

TECHNICAL SKILLS

Robotics: PyTorch, Deep reinforcement learning, ROS1/2, Gazebo, Isaac Sim, ManiSkill, MuJoco.
Programming: Python, C, C++.
Design and tools: SolidWorks, Autodesk Fusion 360.